ALICE L WALTON SCHOOL OF MEDICINE

Aplastic Anemia

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Adapted by: Chelsey Deel
Editor: Jordan Finn
Adapted from: Aplastic Anemia

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Learning Objectives (5) Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of the mentioned anemia.
2. Explain the pathophysiology of the mentioned anemia.
3. Interpret clinical findings to properly diagnose the mentioned anemia.
4. Outline the
5. Distinguish aplastic anemia from pure red cell aplasia.

CASE CONNECTION

TJ, a 32-year-old female, has been referred by her neurologist for a consultation regarding purplish spots of unknown origin on her skin. TJ says she's noticed easy bruising for about 2 weeks, recurrent nosebleeds, and fatigue. She is prescribed carbamazepine for a seizure disorder but takes no other medications. On examination, she is pale, has fresh blood in her anterior nares, and diffuse ecchymoses and petechiae. A complete blood count is ordered stat, and the lab comes back with critical results: her white blood cell count, hemoglobin, and platelet count are all low. No abnormal cells are seen.

What is TJ's diagnosis? What management should be recommended? Consider your answer as you read, and we'll revisit TJ at the end of the brick.

GO TO CONCLUSION ↓

What Is Aplastic Anemia?

Most anemias result from problems confined to the red blood cells (RBCs) such as limited hemoglobin production in iron deficiency anemia, blocked RBC production in chronic kidney disease, or shortened RBC lifespan in hemolytic anemias.

In aplastic anemias, reduced RBC production is caused by a loss of RBC precursors in the bon … tion in circulating RBCs, granulocytes, … **pancytopenia** (*pan-* meaning all, *cyto-* meaning cell, and *-penia* meaning deficiency). So, unlike most anemias, aplastic anemias cause not just low RBC, but reductions in other blood cell types.

The word "aplastic," which comes from the Greek *a-* (meaning without) and *-plasia* (meaning development), is an apt summary of the problem in this disease: **myeloid hematopoietic stem cells (HSCs)** in the bone marrow cease to proliferate, grow, and mature. As a result, the number of HSCs in the bone marrow is markedly decreased and is replaced by adipose tissue. The loss of myeloid precursors causes the low RBCs, platelets, granulocytes, and monocytes. Many factors can cause this loss of HSCs, including medications and toxins.

This is a relatively rare disease: only about 2 new cases of aplastic anemia per million people are diagnosed annually in the United States. The incidence of aplastic anemia is higher in Asian countries. This is most likely due to increased exposure to toxic environmental factors, rather than genetic predisposition, since the higher incidence is not seen in Asian people living in the United States.

How Does Aplastic Anemia Present (LO #1)?

The clinical presentation of aplastic anemia is a direct consequence of pancytopenia:

- Fatigue, dizziness, and shortness of breath from anemia
- Petechiae, easy bruising, and bleeding from low platelets
- Recurrent infections from reduced granulocytes and monocytes

Note that patients may have varying degrees of these symptoms.

What are some presenting symptoms of aplastic anemia?

What Is the Pathophysiology of Aplastic Anemia (LO #2)?

The ultimate mechanism of aplastic anemia irrespective of etiology is loss of HSCs in the bone marrow, specifically of the **myeloid stem cells**. Recall that these give rise to:

- Erythroblasts, which become erythrocytes (RBCs)
- Megakaryoblasts, which become platelets
- Myeloblasts, with become neutrophils, basophils, eosinophils, and mast cells
- Monoblasts, which become monocytes and macrophages

Figure 1 summarizes this development.

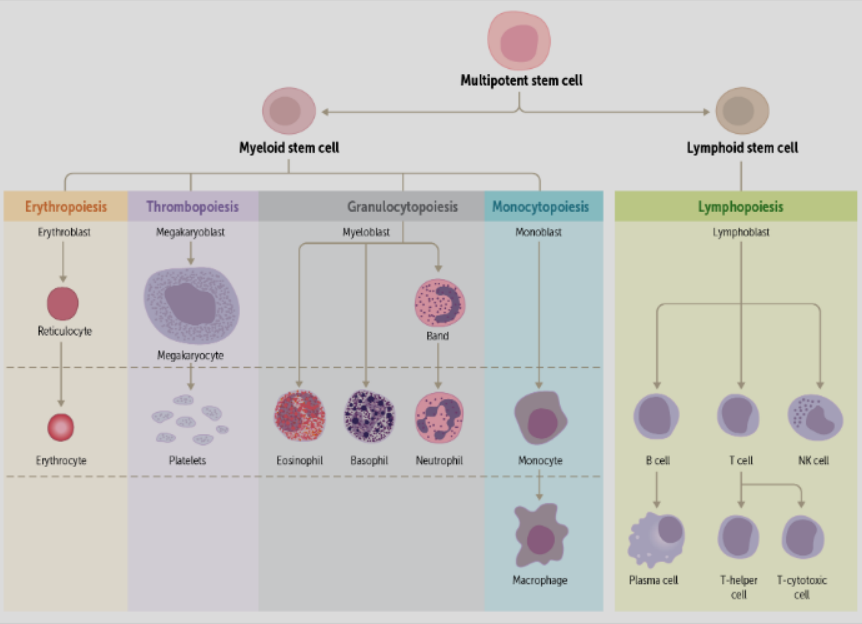

Figure 1

When myeloid HSCs are lost, all cells derived from that line may be decreased, hence the hypocellularity and pancytopenia seen in the bone marrow and the pancytopenia in the peripheral blood.

What is the mechanism underlying all cases of aplastic anemia?

Etiology

Although most cases of aplastic anemia are idiopathic, several conditions and substances can lead to aplastic anemia by damaging HSCs. Of the known causes, drugs and viral infections are the most common, followed by autoimmune, genetic, and hereditary diseases.

Drugs, Toxins, and Radiation. Drugs and toxins that damage the bone marrow are a common cause of aplastic anemia.

CLINICAL CORRELATION

In most cases, the severity of HSC damage by drugs, toxins, and radiation is **dose dependent**. The larger the initial and cumulative dose, the greater the risk. The most common drugs in this category are **chemotherapy drugs**, which are designed to kill rapidly dividing cells and can sometimes kill normal stem cells at the same time, leadi … dent effects are seen with immunosuppressive drugs such as azathioprine and the antibiotic chloramphenicol.

Ionizing radiation used for radiotherapy in cancer also causes dose-dependent toxicity on the bone marrow by generating free radicals. Toxins such as benzene, solvents, and insecticides cause DNA breaks in HSCs, which induce apoptosis.

In other cases, the drug reaction is **idiosyncratic**. These drugs or their metabolites can bind to antigens on stem cells, making them immunogenic and susceptible to T cell–mediated cytotoxicity. Some common examples include:

- Antibiotics: sulfonamides, chloramphenicol
- Anti-epileptics: carbamazepine, valproic acid, and phenytoin
- Methimazole (antithyroid medication)
- Nifedipine
- Nonsteroidal anti-inflammatory drugs (NSAIDs)

Viral Infection. Viruses such as Epstein-Barr, HIV, hepatitis B and C, and herpes can induce HSC destruction in the bone marrow. This occurs either by altering the antigens on blast cells or by activating T cells. Activated cytotoxic T cells become self-reactive against altered antigens and destroy HSCs.

Autoimmune Diseases. Systemic lupus erythematosus and graft vs host disease (after bone marrow transplantation of stem cells) induce autoimmune destruction of HSCs, causing pancytopenia.

Acquired Genetic Mutations. Rarely, patients develop aplastic anemia as part of an acquired gene mutation in the HSCs, often accompanied by another disorder. Examples in … binuria (autoimmune hemolytic anemia, thrombosis, hemoglobinuria) and myelodysplastic syndromes (another cause of bone marrow failure).

Hereditary Conditions. The most common form of the rare hereditary aplastic anemias is **Fanconi anemia**, an autosomal recessive genetic condition in which DNA repair mechanisms are defective. This leads to increased fragility of chromosomes when exposed to things that can damage DNA (i.e. ionizing radiation or chemicals). Accumulated DNA mutations result in apoptosis and destruction of HSCs.

CLINICAL CORRELATION

Fanconi anemia usually presents during childhood (median age ~7 years) with gradually progressive pancytopenia (thrombocytopenia is usually the first abnormality). 2/3 of patients have other constitutional features. Fanconi anemia also causes increased malignancies, hyper- or hypopigmentation of the skin (including cafe-au-lait spots), short stature, dysmorphic facies, and bone abnormalities of the radius and thumb (Figure 2). Eye abnormalities can also be seen, such as strabismus and widely spaced eyes, and structural renal abnormalities like ectopic or horseshoe kidney.

Diagnostic testing includes demonstrating increased chromosomal breaks in lymphocyte culture with agents that cause DNA cross-links (such as diepoxybutane or DEB and mitomycin C).

Some patients may respond to androgens or corticosteroids. Allogeneic stem cell transplant is the treatment of choice for severe cases of Fanconi anemia. It reverses the hematologic abnormalities but not the predisposition to malignancies.

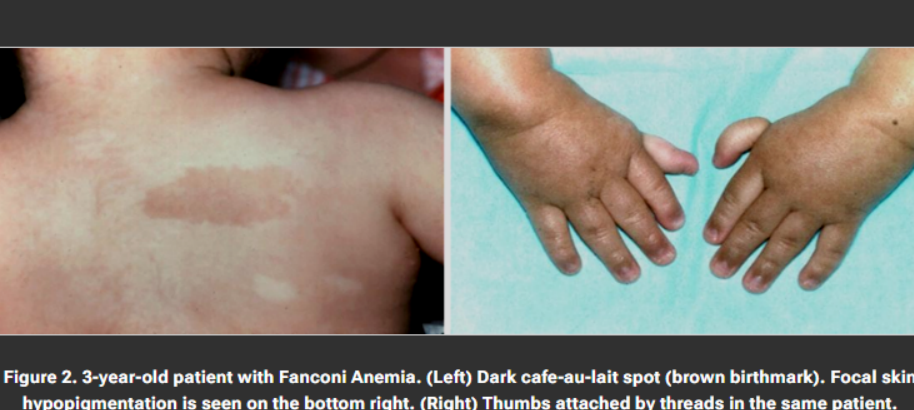

Figure 2. 3-year-old patient with Fanconi Anemia. (Left) Dark cafe-au-lait spot (brown birthmark). Focal skin hypopigmentation is seen on the bottom right. (Right) Thumbs attached by threads in the same patient.

Pathology

In aplastic anemia, the number of myeloid HSCs in the bone marrow is markedly decreased, giving the bone marrow hypocellular morphology (Figure 2, right).

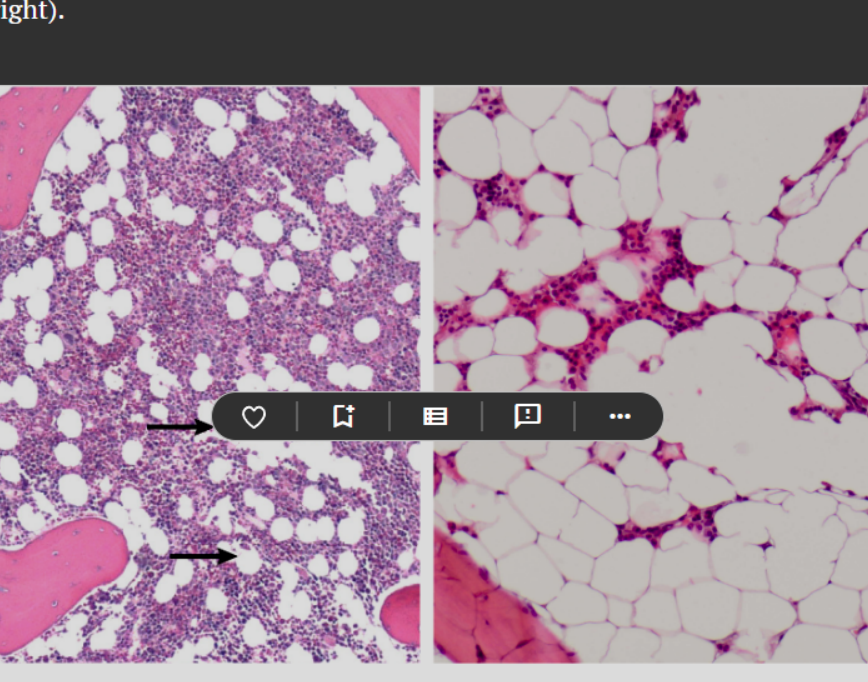

Figure 2

In hypocellular bone marrow, abundant adipocytes (white globules) fill the spaces left by the destroyed stem cells. This contrasts with normal bone marrow (Figure 2, left), where abundant hematopoietic stem cells are interspersed between scant adipose tissue (black arrows), which appear as white globules on H&E stain.

What is the typical appearance of the bone marrow in aplastic anemia?

How Do We Diagnose Aplastic Anemia (LO #3)?

A clinical presentation indicative of **pancytopenia** (e.g., bleeding, infections, fatigue, anemia) should raise the suspicion for aplastic anemia. Combined findings on physical examination, a complete blood count (CBC), and bone marrow biopsy will aid in the diagnosis of aplastic anemia.

Physical Examination

Patients may have signs of anemia such as pallor and tachycardia; signs of thrombocytopenia such as epistaxis, menorrhagia, and bleeding gums; and signs of neutropenia such as fever and sepsis (hypotension and tachycardia).

Laboratory Tests

The initial test when aplastic anemia is suspected is a CBC with differential, which will show low levels of all blood cells (**pancytopenia**) except for lymphocytes, which remain normal because they derive from lymphoid, not myeloid, progenitor cells. The anemia is normocytic (normal mean corpuscular volume). Reticulocytes, which are markers of bone marrow RBC synthesis, will be low. RBCs will appear decreased but with normal morphology on the peripheral smear.

CLINICAL CORRELA…

Aplastic anemia does also affect lymphoid progenitor cells to a lesser degree. Myeloid progenitor cells are the predominant target in these cases.

CLINICAL CORRELATION

Other causes of pancytopenia include disorders in which the bone marrow is infiltrated by neoplastic cells or fibrosis (leukemias, lymphomas, myeloproliferative disorders), bone marrow infection (tuberculosis), nutritional deficiencies (vitamin B_{12}, folate), sequestration of blood cells in the spleen (e.g., hypersplenism due to cirrhosis), or excessive blood cell destruction in the thrombotic microangiopathies (a subset of cases of microangiopathic hemolytic anemia).

Why will lymphocytes be normal on a complete blood count (CBC) in patients with aplastic anemia?

Bone Marrow Biopsy

To confirm the diagnosis, a **bone marrow biopsy** must be performed to evaluate bone marrow morphology and rule out other potentially treatable causes of bone marrow failure such as primary myelofibrosis or metastatic carcinoma. Aplastic anemia will show near-complete replacement of HSCs with adipose tissue (see Figure 2).

How Do We Manage Aplastic Anemia (LO #4)?

Management of aplastic anemia is initially aimed at eliminating or limiting exposure to potential secondary causes, including medications, chemicals, toxins, and ionizing radiation.

Initial treatment is supportive, including transfusions of RBC or platelets and appropriate antibiotics for infections.

Idiopathic cases are treated with medications or stem cell transplantation in the setting of severe transfusion dependence or life-threatening infections.

Medication options include **eltrombopag (EPAG)**. EPAG is a thrombopoietin receptor agonist on stem cells that increases production of all cell lines, not just platelets. The immunosuppressive drugs cyclosporine A or horse antithymocyte globulin are used in severe cases.

Allogenic stem cell transplantation (i.e., from a donor) may be curative, but the mortality rate of the procedure is high, so it is only considered after other treatments fail. Comp… s and graft vs host disease, in which transplanted cells attack the recipient.

When is aplastic anemia treated with drugs or stem cell transplantation?

Prognosis

Recent advances in treatment modalities have greatly improved the prognosis in patients with aplastic anemia. Currently, the 5- to 10-year survival rate is more than 80% in treated patients, especially those who are young with mild-to-moderate disease. In contrast, as high as 7 out of 10 untreated patients will die in 1 year, with the majority being elderly patients and those with severe disease.

When untreated, which patient demographic group is most at risk of death from aplastic anemia?

Pure Red Cell Aplasia (LO #5)

Pure red cell aplasia (PRCA) is anemia with decreased erythroid precursors in the bone marrow; granulocytes, megakaryocytes/platelets are not affected. Like aplastic anemia, there are both inherited/congenital and acquired forms.

The most important congenital form of PRCA is **Diamond-Blackfan syndrome** (also known as Diamond-Blackfan anemia and congenital hypoplastic anemia). This is a rare inherited bone marrow failure syndrome, associated with mutations in genes that encode ribosomal proteins, characterized by macrocytic anemia, reticulocytopenia (decreased reticulocytes), and cancer predisposition (leukemia, myelodysplastic neoplasms, osteosarcoma). Other congenital abnormalities may be present (skeletal, cardiac) (Figure 3).

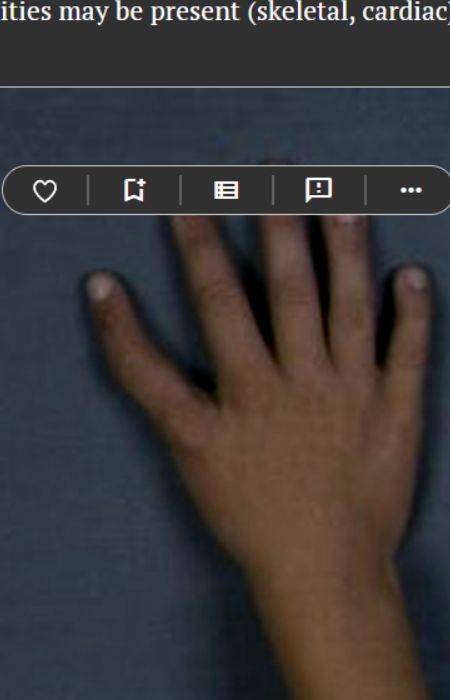

Figure 3. Triphalangeal thumb in a patient with Diamond-Blackfan syndrome. A triphalangeal thumb is a rare congenital anomaly where the thumb has three phalanges (the bones the make up the fingers and thumb) instead of the usual two. This can cause the thumb to appear longer than normal, potentially affecting its function and appearance.

CLINICAL CORRELATION

Most children with Diamond-Blackfan syndrome experience symptoms very early in life, usually before their first birthday, but it can appear at any age. The most common symptoms are pale skin, fatigue (infants may get tired during feeding), and dyspnea.

The majority of cases of Diamond-Blackfan syndrome respond to corticosteroids, with … cell transplant as other options (only transplant is curative).

Acquired causes of pure red cell aplasia can be associated with malignancies (such as thymoma, chronic lymphocytic leukemia), autoimmune disorders, medications, infection (such as parvovirus B19), or idiopathic (no obvious associated disease or precipitating factor).

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to TJ, what is her diagnosis? What management should be recommended?

TJ has pancytopenia that is likely drug-induced aplastic anemia. Her low platelet count explains her bruising, nosebleeds, and petechiae, and her low hemoglobin explains her fatigue. She has not yet had infections due to her low white blood cell count. Her neurologist is notified, and a recommendation is made to replace the causative drug, carbamazepine, with another approved for her seizure disorder. You tell TJ that her medication is likely the cause of her low cell counts and that with discontinuation of the medication, her counts should rebound to normal. In the meantime, you tell her that she requires close follow-up to evaluate her need for transfusions or marrow-stimulating factors.

Summary

- Aplastic anemia occurs when myeloid hematopoietic stem cells in the bone marrow are decreased.
- The reduction in circulating red blood cells (RBCs), granulocytes, monocytes, and p
- Patients may present with symptoms caused by pancytopenia, including those from anemia (eg, fatigue, dizziness, and shortness of breath), low platelets (eg, petechiae, easy bruising, and bleeding), and low granulocytes and monocytes (eg, recurrent infections).
- Patients may have varying degrees of these symptoms.
- Most cases of aplastic anemia are idiopathic.
- Some cases have known causes such as chemotherapy, idiosyncratic drug reactions, toxic chemicals, viral infections, radiation, and acquired/hereditary gene mutations.
- The most common form of the rare hereditary aplastic anemias is Fanconi anemia, an autosomal recessive genetic condition in which DNA repair mechanisms are defective. It usually presents during childhood with gradually progressive pancytopenia. 2/3 of patients have other constitutional features
- Patients may have signs and complete blood count (CBC) findings of pancytopenia.
- A bone marrow biopsy will show near-complete loss of hematopoietic stem cells with abundant adipose tissue.
- Treatment involves removal of any known causes, support with transfusions, stimulation of cell production with a thrombopoietin receptor agonist, or immunosuppressive drugs.
- Allogenic stem cell transplantation is used in the most severe cases.
- The 5- to 10-year survival rate is very high in treated patients.
- Pure red cell aplasia (PRCA) is anemia with decreased erythroid precursors in the bone marrow; granulocytes, megakaryocytes/platelets are not affected. There are both congenital and acquired causes.
- The most important congenital form of PRCA is **Diamond-Blackfan syndrome** (also … and congenital hypoplastic anemia). This is a rare inherited bone marrow failure syndrome characterized by macrocytic anemia, reticulocytopenia (decreased reticulocytes), and cancer predisposition (leukemia, myelodysplastic neoplasms, osteosarcoma). Other congenital abnormalities may be present (skeletal, cardiac).
- Acquired causes of pure red cell aplasia can be associated with malignancies (such as thymoma, chronic lymphocytic leukemia), autoimmune disorders, medications, infection (such as parvovirus B19), or idiopathic (no obvious associated disease or precipitating factor).

Review Questions

1. Which of the following bone marrow findings is most likely in a patient with aplastic anemia?

A. ○ Bone marrow cellularity of 30%
B. ○ Extensive bone marrow fibrosis
C. ○ Large lymphoid aggregates
D. ○ Marked proliferation of erythroid precursors
E. ● Replacement of marrow with adipose tissue

Hide Explanation

The correct answer is replacement of marrow with adipose tissue (E). In aplastic anemia, the number of hematopoietic stem cells is decreased in the bone marrow, which is filled with adipose tissue. Bone marrow cellularity of 30% (A) would be a normal finding in a 78-year-old patient because of the normal age-related decline. Extensive bone marrow fibrosis (B) might explain this patient's pancytopenia, but it's not diagnostic for aplastic anemia. Large lymphoid aggregates (C) and marked proliferation of erythroid precursors (D) are seen in myeloproliferative disorders but not in aplastic anemia.

2. Which of the following statements is most accurate regarding the mechanism of aplastic anemia?

A. ○ Benzene activates fibroblasts
B. ○ Drugs activate T cells
C. ● Radiation generates toxic free radicals
D. ○ Viruses alter T-cell antigen
E. ○ Vitamin deficiency inhibits cell division

Hide Explanation

The correct answer is radiation generates toxic free radicals (C). Ionizing radiation directly damages hematopoietic stem cells (HSCs) by generating free radicals within the cells. Benzene ac… benzene is an alkylating agent that cross-links and breaks DNA in HSCs, which induces apoptosis. Drugs activate T cells (B) is incorrect because drugs act as haptens that make HSC antigens immunogenic after binding to them. This results in T-cell mediated destruction of HSCs. Drugs do not directly activate T cells, they alter HSC antigens. Viruses alter T-cell antigen (D) is incorrect because viruses alter the antigens of infected HSCs or activate infected T cells. Altered HSCs are prone to activated T cell–mediated cytotoxicity. Vitamin deficiency inhibits cell division (E) is incorrect, because, although it does inhibit cell division and cause anemia, it would not cause aplastic anemia.

3. Which of the following is the least likely complication of aplastic anemia?

A. ○ Dyspnea
B. ○ Menorrhagia
C. ○ Pharyngitis
D. ○ Syncope
E. ● Thrombosis

Hide Explanation

The correct answer is thrombosis (E). Increased clot formation occurs in hypercoagulable states such as protein C or S deficiency, or in malignancy. Patients with aplastic anemia h… form clots. The other conditions could all b… pnea (A) is a manifestation of severe anemia or pulmonary infection. Menorrhagia (B) is a manifestation of low platelet count. Pharyngitis (C) and other infections may increase due to neutropenia. Syncope (D) is due to low cerebral oxygen delivery that occurs in any form of severe anemia.

4. After returning from a camping trip in North Carolina, a 19-year-old female develops fever, myalgia, headache, and a rash on her wrists and ankles. She is diagnosed with Rocky Mountain spotted fever and treated with chloramphenicol because she is allergic to tetracycline. A week later, she returns with complaints of worsening fatigue and mucosal bleeding. A complete blood count reveals marked decreases in red and white blood cells and platelets. Which of the following is the best next step in management of this patient?

A. ○ Administer N-acetylcysteine
B. ○ Decrease the dose of chloramphenicol
C. ● Discontinue chloramphenicol
D. ○ Perform stem cell transplantation
E. ○ Administer granulocyte colony-stimulating factor

Hide Explanation

The correct answer is … patient began treatment with chloramphenicol, and then developed signs and symptoms consistent with aplastic anemia. Although a definitive diagnosis of aplastic anemia was not made here, a strong clinical suspicion in the presence of a known causative agent should prompt immediate discontinuation of the drug. Administer N-acetylcysteine (A) is incorrect, as that agent is used for acetaminophen overdose. Decrease the dose of chloramphenicol (B) is insufficient, given the life-threatening nature of aplastic anemia. Perform stem cell transplantation (D) is not the best choice, as that is used as a last resort for idiopathic aplastic anemia, after more conservative measures have failed. Administer granulocyte colony-stimulating factor (E) is most typically done in neutropenic fever, especially after chemotherapy for cancer, but not usually in aplastic anemia. In any case, withdrawing the causative drug should occur first.

5. A 5-year-old patient presents to the clinic with easy bruising. The parents are also concerned with their lack of growth in height. Physical examination reveals a patient with short stature for their age, bruises and petechiae on their skin, and occasional cafe-au-lait spots. Laboratory testing reveals pancytopenia. What is the next best diagnostic step?

A. ○ Bone marrow biopsy
B. ● Chromosomal breakage test
C. ○ Hepatitis testing
D. ○ HIV testing
E. ○ Parvoviru

Hide Explanation

The correct answer is chromosomal breakage test (B). The clinical picture is suspicious for Fanconi anemia. One of the diagnostic tests for Fanconi anemia includes demonstrating increased chromosomal breaks in lymphocyte culture with agents that cause DNA cross-links (such as diepoxybutane or DEB and mitomycin C). Bone marrow biopsy (A) would show aplastic anemia but would not specifically establish the diagnosis of Fanconi anemia. Hepatitis testing (C) and HIV testing (D) are important tests in the evaluation of aplastic anemia; however, the clinical symptoms and patient age put Fanconi anemia at the top of the differential. Parvovirus testing/serology (E) would be useful in patients with pure red cell aplasia as a result of Parvovirus B19 infection; CBC would show anemia, not pancytopenia.

How would you rate this Brick?

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Check off the objectives when you feel comfortable with the concept

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Objective Confidence: 5/5